

MH1101 Tutorial 6 (Week 7)

Reference: Sections 3.4, 3.5, 3.6 (Lecture Notes)

1. Evaluate the integral.

(i) $\int_1^2 \frac{x^3 + 4x^2 + x - 1}{x^3 + x^2} dx$

(ii) $\int \frac{4x}{x^3 + x^2 + x + 1} dx$

(iii) $\int \frac{x + 4}{x^2 + 2x + 5} dx$

(iv) $\int \frac{5x^4 + 7x^2 + x + 2}{x(x^2 + 1)^2} dx$

(v) $\int \frac{e^{2x}}{e^{2x} + 3e^x + 2} dx$

2. Let T_n and M_n be the approximations to the integral $\int_0^1 \cos(x^2) dx$ using Trapezoidal Rule and Midpoint Rule respectively.

- (a) Find the approximation T_8 and M_8 .
- (b) Estimate the errors in the approximations of part (a).
- (c) How large do we have to choose n so that the approximations T_n and M_n to the integral in part (a) are accurate to within 0.0001?

3. Let S_n be the approximation of the integral $\int_1^5 \ln x dx$ using Simpson's Rule.

- (i) Calculate S_8 .
- (ii) Find a value n such that S_n has error of at most 10^{-6} .

4. Show that if $f(x) = px^2 + qx + r$ is a quadratic polynomial, then $S_2 = \int_a^b f(x) dx$, i.e. Simpson's Rule with $n = 2$ gives the exact value of $\int_a^b f(x) dx$.

Answer Keys. **1.** (i) $\frac{1}{2} + \ln 6$ (ii) $-2 \ln |x+1| + \ln |x^2+1| + 2 \tan^{-1} x + C$
 (iii) $\frac{1}{2} \ln |x^2+2x+5| + \frac{3}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C$ (iv) $2 \ln |x| + \frac{3}{2} \ln |x^2+1| + \frac{\tan^{-1} x}{2} + \frac{x}{2(x^2+1)} + C$ (v) $\ln \frac{(e^x+2)^2}{e^x+1} + C$ **2.** (a) $T_8 \approx 0.902333$, $M_8 \approx 0.905620$ (c) $n = 71$ for T_n , $n = 50$ for M_n **3.** (i) $S_8 = 4.046655$ (ii) $n = 78$